Yuxiang Lin

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Education

Duke Kunshan University (DKU) & Duke University Dual Degree, Class of 2026

Duke Kunshan University **Kunshan, China**

B.S. in Applied Math & Computational Science; Computer Science

- GPA: 3.962/4.0; Dean's List with Distinction (top 10%), 4 semesters (Fall 2022–Spring 2024); Dean's List (top one-third) Fall 2024
- Scholarships: Hong Scholarship (admission), 2023 National Encouragement Scholarship (9 recipients in DKU)
- Courses: MATH 305 (Linear Algebra, A), MATH 307 (Complex Analysis, A), COMPSCI 308 (Algorithms, A+)

Duke University **Durham, U.S.**

B.S. in Interdisciplinary Studies; Applied Math & Computational Science; Computer Science

- Courses: COMPSCI 553 (Compiler Construction, A+), COMPSCI 650 (Parallel Computer Architecture)

Research Experience

Prof. Danfeng Zhang (Duke CS+ Summer Program) **Durham, U.S.**

Researcher in Program Verification and Software Reliability *05/2025–Present*

- Modified *KLEE* source code to verify relational properties and analyze input-dependent performance variations
- Generated concrete inputs to demonstrate divergent execution paths across paired program runs
- Achieved 10× speedup over prior verification techniques using product-program construction
- Developed an automated benchmarking suite to verify reported property violations
- Refuted six human-verified software bugs in peer-reviewed work as false positives
- Revealed control-flow dependencies among software defects, a factor overlooked in prior studies

Prof. Italo Simonelli (Undergraduate Signature Work) **Kunshan, China**

Researcher in Matroids and Sunflower Lemma *01/2024–Present*

- Proposed *weak matroids*, a new combinatorial structure obtained by relaxing the matroid augmentation axiom
- Applied weak matroids to digraphs (graphic WMs), establishing that acyclic edge sets are independent
- Remarked that graphic WMs guarantee a minimum feedback arc set independent of the edge selection order
- Formulated *minimum sunflower-critical family* (*MSCF*), a variant of the sunflower lemma
- Defined MSCF as the smallest sunflower-free family that form a sunflower after adding any other set
- Utilized C++ and the Boost Graph Library for high-performance combinatorial enumeration experiments

DKU Edge Intelligence Lab (Prof. Bing Luo, DKU Summer Research Scholars) **Kunshan, China**

Lead Developer for Campus Information System *05/2024–05/2025*

- Led development of *ChatDKU*, an open-source information assistant for the DKU community
- Implemented a structured retrieval-augmented generation (RAG) architecture using LlamaIndex and DSPy
- Authored 26,000+ lines of code and contributed upstream fixes to LlamaIndex and DSPy
- Customized frameworks for direct prompt control, significantly improving response quality

Academic Contest

DKU Programming Contest Club **Kunshan, China**

Member of DKU ICPC Teams *10/2022–11/2024*

- Silver Medal (Rank 108/368) at the 2023 ICPC Asia Xi'an Regional Contest
- Bronze Medal (Rank 149/281) at the 2022 ICPC Asia-East Continent Final Contest
- Silver Medal (Rank 100/417) at the 2022 ICPC Asia Nanjing Regional Contest
- Gold Medal (Rank 20/452) at the 2022 ICPC Asia Hefei Regional Contest

Projects & Presentations

DKU Innovation Incubator **Kunshan, China**

Founder and Leader of the Affordable Embroidery Machine Project *09/2023–05/2025*

- Received Certificate of Completion for College Student Innovation and Entrepreneurship Training Program
- Designed an affordable and scalable CNC embroidery machine using FreeCAD
- Constructed a 3D-printed prototype and implemented motion control using Arduino

DKU Edge Intelligence Lab (Prof. Bing Luo) **Kunshan, China**
Backend Developer of FedCampus *01/2024–04/2024*
- Maintained the Django backend for a federated learning and analytics platform for DKU with 90 active users

Publications

- (**ICLR 2026, Accepted**) Shao, J., **Lin, Y.**, Lohani, M. P., Miao, Y., and Luo, B., "Do LLM Agents Know How to Ground, Recover, and Assess? A Benchmark for Epistemic Competence in Information-Seeking Agents", arXiv e-prints, arXiv:2509.22391, 2025.
  - Addressed the limitation of existing answer-only evaluations that ignore reasoning flaws
  - Developed a process-level evaluation framework to assess a model's *epistemic competence*

Teaching & Mentoring

Prof. Jiang Long **Kunshan, China**
COMPSCI 201 (Introduction to Programming and Data Structures) Teaching Assistant *08/2023–12/2023*
- Provided in-depth technical support to 20 students during weekly office hours
- Reinforced course concepts including Java object-oriented programming and automata theory

DKU Academic Resource Center **Kunshan, China**
STATS 102 (Introduction to Data Science) Peer Tutor *08/2023–05/2024*
- Trained for 11 hours in *Tutor Essentials*, a CRLA-aligned tutor training course
- Conducted bi-weekly tutoring sessions to assist students with Python data science workflows using scikit-learn

Leadership & Service

DKU Programming Contest Club **Kunshan, China**
President, Vice President, and Marketing Coordinator *10/2022–12/2024*
- The club received 2023 DKU Academic Student Organization of the Year Award
- I received 2023 DKU Student Leader of the Year Award and 2023 DKU Community Service Award
- Co-founded DKU Programming Contest Club with 100 members and hosted more than 30 seminars on algorithms
- Led the club members to win 7 medals in ICPC, including 3 gold medals

DKU Metal Club **Kunshan, China**
Secretary and Vice President *10/2022–12/2024*
- Co-founded DKU Metal Club, led recruitment for 160 members
- Coordinated the Iridodonesis Debut metal music show with 40 attendees

Additional Information

Languages: Chinese Mandarin (Native), English (Proficient)

Skills: C++ (used extensively over nine years of competitive programming), Python (used extensively for ChatDKU and other projects), Standard ML (used for compiler construction), Java, Julia, SQL, GNU/Linux, Bash, Git, OpenGL, LaTeX, CAD modeling

Interests: Heavy metal music, guitar, motorcycling, bicycling, basketball, swimming, video games